Chapter 12: Binomial and Poisson Distributions

The Practice of Statistics in the Life Sciences
by Brigitte Baldi and David S. Moore

Instructor: Ali Raisolsadat

Part II: Probability and Distributions

Why This Chapter Matters

The previous chapter focused on continuous measurements. This chapter turns to **counts**.

Examples of counts include the number of type O blood donations in a short sequence of donors, the number of contaminated containers in a sample, the number of tagged fish in a catch and the number of disease cases in a month.

Two important count models will be developed.

- The **binomial distribution** for successes among a fixed number of trials
- The **Poisson distribution** for event counts in a fixed interval of time, space or exposure

Introduction: Counting Successes

A **count** records how many times a specified outcome occurs. Examples include:

- How many of 800 students have perfect vision?
- How many of 250 patients meet a study's five-year outcome?
- How many of 10 trees survive the winter?
- How many of 50 seeds germinate within two weeks?
- How many of 100 manufactured sensors pass a calibration test?

Before calculating

We first decide whether the process producing the count matches a known probability model. A "success" simply means the outcome we are counting.

The Binomial Setting

A binomial setting has four conditions.

The Four Conditions

1. Two outcome categories labelled success and failure (**B**inary outcomes)
2. **I**ndependent trials denoted by k
3. fixed **N**umber of trials denoted by n
4. **S**ame success probability on every trial denoted by p

The word **success** is only a label. It can refer to a desirable outcome or an undesirable one such as a contamination event.
When these conditions are satisfied, the count X of successes has a binomial distribution.

$$X \sim \text{Bin}(n, p)$$

The possible values are the whole numbers from 0 through n.

Example 12.1: Blood Donations

Question

Blood type O has probability 0.45 in the population used for this example. The next five donations come from unrelated individuals. Does the number of type O donations have a binomial distribution?

Solution

The number of donations is fixed at five. Each donation is classified as type O or not type O. The success probability is modelled as 0.45 on every trial. The donors are unrelated, so the outcomes are modelled as independent.

All four binomial conditions are satisfied. If X is the number of type O donations, then

$$X \sim \text{Bin}(5, 0.45)$$

Example 12.2: Selecting Volunteer Subjects

Question

A researcher has 40 volunteers consisting of 20 men and 20 women. Ten volunteers are selected without replacement. Does the number of men selected have a binomial distribution?

Solution

The sample size is fixed at 10 and each selection can be classified as man or woman. The difficulty is sampling without replacement. After one volunteer is selected, the composition of the pool changes.

The trials are not independent and the success probability is not constant. The exact count is therefore not binomial.

Example 12.3: A Large Population Approximation

Question

A pharmaceutical company inspects 10 plastic containers selected without replacement from a shipment of 10,000 containers. Suppose 10% of the shipment is contaminated. Can the number of contaminated containers be approximated by a binomial distribution?

Solution

Sampling without replacement is not exactly independent. In this example, however, the sample represents only 0.1% of the population. Removing one container has a negligible effect on the contamination probability for the next draw.

The 10% condition is easily satisfied, so a binomial approximation is very accurate.

$$X \approx \text{Bin}(10, 0.10)$$

From a Count to a Probability

A binomial probability answers a question about the number of successes among a fixed number of trials.

The probability of exactly k successes can be understood in two steps.

1. Find the probability of one specific arrangement of successes and failures
2. Count how many arrangements produce the same total number of successes

Example 12.4: Blood Donations

Question

Each donor has probability 0.45 of having type O blood. Among the next five unrelated donors, what is the probability that exactly two have type O blood?

Explanation

The count is binomial with $n = 5$ and $p = 0.45$. Exactly two successes means two type O donations and three non-type O donations.

Let S denote the event of having blood type O and F denote the event of **not** having blood type). Then there are 10 possible outcomes.

$$\Omega = \{SSFFF, SFSFF, SFFSF, SFFFS, FSSFF, FSFSF, FSFFS, FFSSF, FFSFS, FFFSS\}$$

Example 12.4: Blood Donations

One particular arrangement is SFSFF.

Because the trials are independent, the probability of this one arrangement is found by multiplication.

$$P(\text{SFSFF}) = (0.45)(0.55)(0.45)(0.55)(0.55)$$

This simplifies to

$$P(\text{SFSFF}) = (0.45)^2(0.55)^3$$

Since every arrangement containing two successes and three failures has this same probability and there are 10 different ways to place two successes among five trials. The overall probability is therefore

$$P(X = 2) = 10(0.45)^2(0.55)^3 = 0.3369$$

The probability that exactly two of the five donors have type O blood is about 0.3369.

Example 12.4: Blood Donations

The exact binomial probability can be calculated with `dbinom()`.

```
# Exact binomial probability for two type O donations
prob_blood <- dbinom(x = 2, size = 5, prob = 0.45)
print(round(prob_blood, 4))
```

```
> [1] 0.3369
```

The output is 0.3369, which matches the calculation from the binomial formula.

The Binomial Coefficients

The binomial coefficient counts how many arrangements contain exactly k successes among n trials.

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Factorial notation is defined by

$$n! = n(n-1)(n-2)\cdots 1$$

with the convention

$$0! = 1$$

Binomial Probability Formula

If $X \sim \text{Bin}(n, p)$, then the probability of exactly k successes is

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

where $k = 0, 1, 2, \ldots, n$.

The coefficient counts arrangements, p^k accounts for the successes and $(1-p)^{n-k}$ accounts for the failures.

Translating Wording into Notation

The wording of a question should be translated before R is used.

- Exactly 2 means $P(X = 2)$
- At most 2 or no more than 2 means $P(X \leq 2)$
- More than 2 means $P(X > 2)$
- At least 2 means $P(X \geq 2)$

This translation step is often more important than the mechanics of the calculation.

Calculating Binomial Probabilities with R

Two R functions are especially important.

Probability question	R function	Meaning
Exactly x successes	`dbinom(x, size, prob)`	Probability mass at one count
At most q successes	`pbinom(q, size, prob)`	Cumulative probability from 0 through q

For discrete random variables, `dbinom()` gives a probability mass rather than a continuous density.

Example 12.5: Inspecting Pharmaceutical Containers

Question

The number of contaminated containers is modelled by $X \sim \text{Bin}(10, 0.10)$. What is the probability that a sample contains no more than one contaminated container?

Explanation

The phrase *no more than one* means the count can be 0 or 1. These outcomes are disjoint, so their probabilities can be added.

$$P(X \leq 1) = P(X = 0) + P(X = 1)$$

Example 12.5: Inspecting Pharmaceutical Containers

Using the binomial formula,

$$P(X \leq 1) = \binom{10}{0}(0.1)^0(0.9)^{10} + \binom{10}{1}(0.1)^1(0.9)^9 = 0.3487 + 0.3874$$

$$P(X \leq 1) = 0.7361$$

Under this model, about 73.61% of samples contain no more than one contaminated container.

Example 12.5: Inspecting Pharmaceutical Containers

The cumulative probability can be found directly with `pbinom()` or by adding two exact binomial masses.

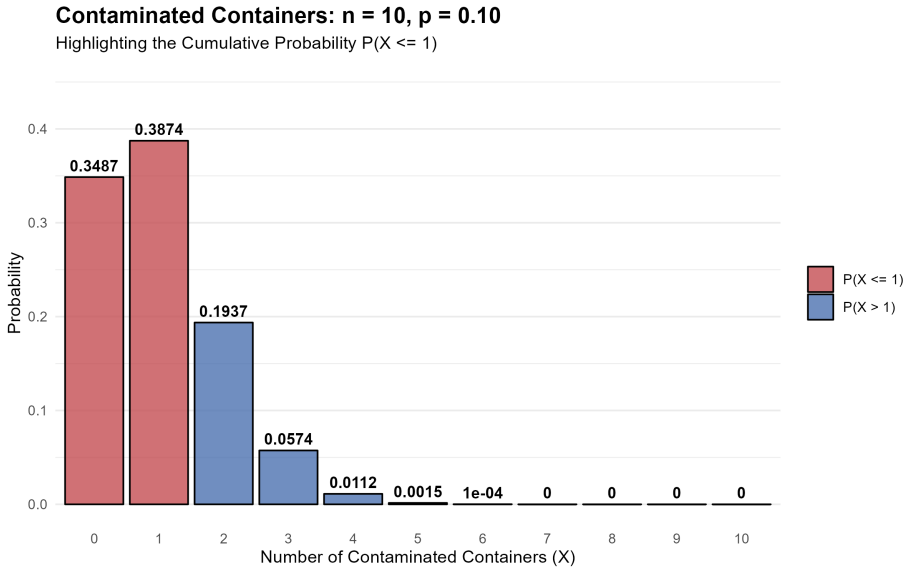
```
# P(X <= 1) for n = 10 and p = 0.10
prob_pharma_cumulative <- pbinom(q = 1, size = 10, prob = 0.10)

prob_pharma_manual <-
  dbinom(x = 0, size = 10, prob = 0.10) + dbinom(x = 1, size = 10, prob = 0.10)

print(round(prob_pharma_cumulative, 4))
> [1] 0.7361

print(round(prob_pharma_manual, 4))
> [1] 0.7361
```

Example 12.5: Inspecting Pharmaceutical Containers



Mean and Variance

A binomial distribution has simple formulas for its centre and spread.

Binomial Mean, Variance and Standard Deviation

If $X \sim \text{Bin}(n, p)$, then

$$\mu = np$$

$$\sigma^2 = np(1-p)$$

$$\sigma = \sqrt{np(1-p)}$$

These formulas are specific to the binomial model.

- The mean is a long-run average across many repetitions of the same binomial setting.
- The standard deviation describes the typical amount of sample-to-sample variation around that mean.
- The mean does not need to be a possible observed count. A mean of 13.5 does not imply that 13.5 successes can occur in one sample.

Example 12.6: Inspecting Pharmaceutical Containers

Question

For $X \sim \text{Bin}(10, 0.10)$, what are the mean and standard deviation of the number of contaminated containers?

Example 12.6: Inspecting Pharmaceutical Containers

Question

For $X \sim \text{Bin}(10, 0.10)$, what are the mean and standard deviation of the number of contaminated containers?

Solution

The mean is

$$\mu = np = (10)(0.10) = 1$$

The variance is

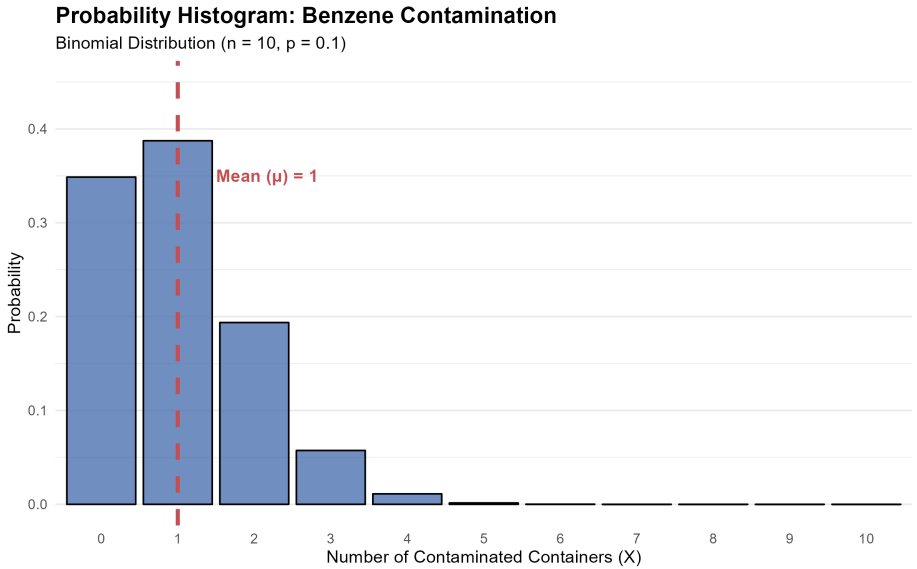
$$\sigma^2 = np(1-p) = (10)(0.10)(0.90) = 0.9$$

The standard deviation is

$$\sigma = \sqrt{0.9} = 0.9487$$

Across many samples of 10 containers, the long-run average number contaminated is 1.

Example 12.6: Inspecting Pharmaceutical Containers



Practice Examples: Binomial Probability, Mean and Variance

The remaining binomial examples use the same pattern.

1. Identify the binomial model
2. Determine whether the question asks for a mean, a variance, a standard deviation or a probability
3. Use the appropriate formula or exact binomial probability
4. Interpret the result in the context of the problem

Practice Example: Genetics

Question

A Mendelian genetics model gives probability 0.25 that a pea-seed cross produces a yellow seed. Four seeds are considered. What are the expected number of yellow seeds, the variance and the probability of exactly one yellow seed?

Practice Example: Genetics

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A Mendelian genetics model gives probability 0.25 that a pea-seed cross produces a yellow seed. Four seeds are considered. What are the expected number of yellow seeds, the variance and the probability of exactly one yellow seed?

Solution

With $X \sim \text{Bin}(4, 0.25)$

$$\mu = (4)(0.25) = 1$$

$$\sigma^2 = (4)(0.25)(0.75) = 0.75$$

$$P(X = 1) = \binom{4}{1}(0.25)(0.75)^3 = 0.4219$$

Practice Example: Genetics

The exact probability is calculated with `dbinom()`.

```
# Probability of exactly one yellow seed
prob_lvl1 <- dbinom(x = 1, size = 4, prob = 0.25)
print(round(prob_lvl1, 4))
```

```
> [1] 0.4219
```

Practice Example: Clinical Medicine

Question

A treatment cures an infected patient with probability 0.80. Ten patients are treated. What are the mean and standard deviation of the number cured? What is the probability that exactly eight patients are cured?

Practice Example: Clinical Medicine

Question

A treatment cures an infected patient with probability 0.80. Ten patients are treated. What are the mean and standard deviation of the number cured? What is the probability that exactly eight patients are cured?

Solution

With $X \sim \text{Bin}(10, 0.80)$

$$\mu = (10)(0.80) = 8$$

$$\sigma = \sqrt{(10)(0.80)(0.20)} = 1.2649$$

$$P(X = 8) = \binom{10}{8}(0.80)^8(0.20)^2 = 0.3020$$

Practice Example: Clinical Medicine

The exact probability is calculated with `dbinom()`.

```
# Probability of exactly eight cured patients
prob_lvl2 <- dbinom(x = 8, size = 10, prob = 0.80)
print(round(prob_lvl2, 4))
```

```
> [1] 0.3020
```

Practice Example: Ecological Sampling

Question

Marine biologists estimate that 30% of a large fish population is tagged. A fisher catches 15 fish. What is the variance of the number of tagged fish? What is the probability of exactly five tagged fish?

Practice Example: Ecological Sampling

Question

Marine biologists estimate that 30% of a large fish population is tagged. A fisher catches 15 fish. What is the variance of the number of tagged fish? What is the probability of exactly five tagged fish?

Solution

With $X \sim \text{Bin}(15, 0.30)$

$$\sigma^2 = (15)(0.30)(0.70) = 3.15$$

$$P(X = 5) = \binom{15}{5}(0.30)^5(0.70)^{10} = 0.2061$$

Practice Example: Ecological Sampling

The exact probability is calculated with `dbinom()`.

```
# Probability of exactly five tagged fish
prob_lvl3 <- dbinom(x = 5, size = 15, prob = 0.30)
print(round(prob_lvl3, 4))
```

```
> [1] 0.2061
```

Practice Example: Expected Count and Exact Count

Question

A medical supply company has a 5% defect rate for glass micropipettes. A laboratory buys a box of 20 micropipettes. What is the expected number of defective micropipettes? What is the probability that the observed count is exactly equal to that expected value?

Practice Example: Expected Count and Exact Count

Question

A medical supply company has a 5% defect rate for glass micropipettes. A laboratory buys a box of 20 micropipettes. What is the expected number of defective micropipettes? What is the probability that the observed count is exactly equal to that expected value?

Solution

With $X \sim \text{Bin}(20, 0.05)$

$$\mu = (20)(0.05) = 1$$

$$P(X = 1) = \binom{20}{1}(0.05)(0.95)^{19} = 0.3774$$

Practice Example: Expected Count and Exact Count

The probability of observing exactly the expected count is

```
# Probability that the observed count equals the mean of one
prob_lvl4 <- dbinom(x = 1, size = 20, prob = 0.05)
print(round(prob_lvl4, 4))

> [1] 0.3774
```

Practice Example: The Reverse Engineer

Question

A vaccine study models the number of volunteers with a mild side effect as binomial with $n = 100$. The success probability p is unknown and $p < 0.50$. The variance of the count is 16. What is p? What is the mean count? What is the probability that exactly 20 volunteers experience the side effect?

Practice Example: The Reverse Engineer

The variance equation is

$$100p(1-p) = 16$$

Rearranging gives

$$25p^2 - 25p + 4 = 0$$

The two mathematical solutions are $p = 0.20$ and $p = 0.80$. The condition $p < 0.50$ gives

$$p = 0.20$$

Then

$$\mu = (100)(0.20) = 20$$

$$P(X = 20) = \binom{100}{20}(0.20)^{20}(0.80)^{80} = 0.0993$$

Practice Example: The Reverse Engineer

After solving for $p = 0.20$, the exact probability is

```
# Probability of exactly 20 side effects after solving for p
prob_lvl5 <- dbinom(x = 20, size = 100, prob = 0.20)
print(round(prob_lvl5, 4))

> [1] 0.0993
```

Practice Example: Vaccine Efficacy

Question

A seasonal flu vaccine is effective with probability 0.85 for each of 25 patients. What are the mean, variance and standard deviation of the number protected? What is the probability that exactly 22 patients are protected?

Practice Example: Vaccine Efficacy

Question

A seasonal flu vaccine is effective with probability 0.85 for each of 25 patients. What are the mean, variance and standard deviation of the number protected? What is the probability that exactly 22 patients are protected?

Solution

With $X \sim \text{Bin}(25, 0.85)$

$$\mu = (25)(0.85) = 21.25$$

$$\sigma^2 = (25)(0.85)(0.15) = 3.1875$$

$$\sigma = \sqrt{3.1875} = 1.7854$$

$$P(X = 22) = \binom{25}{22}(0.85)^{22}(0.15)^3 = 0.2174$$

Practice Example: Vaccine Efficacy

The mean, variance, standard deviation and exact probability can all be calculated in R.

```
n <- 25
p <- 0.85
mean_calc <- n * p
var_calc <- n * p * (1 - p)
sd_calc <- sqrt(var_calc)
prob <- dbinom(x = 22, size = n, prob = p)

print(round(mean_calc, 4))
> [1] 21.25
print(round(var_calc, 4))
> [1] 3.1875
print(round(sd_calc, 4))
> [1] 1.7854
print(round(prob, 4))
>[1] 0.2174
```

Break Time



From Fixed Trials to Events in an Interval

The **binomial model** counts successes among a fixed number of trials.
The **Poisson model** counts events within a fixed interval of time, space, or exposure. Possible applications include:

- **Time:** the number of calls an animal rescue centre receives during one hour.
- **Area:** the number of seedlings found in a $1\,\text{m}^2$ sampling plot.
- **Length:** the number of manufacturing defects along one metre of fabric.
- **Exposure:** the number of radioactive decays detected during a 10-second observation.

When is a Poisson model reasonable?

Events occur independently at an approximately constant rate over the interval. Clustering or changing rates can make the model unsuitable.

The Poisson Setting

A Poisson model is based on assumptions about how events occur within the interval.

The Poisson Setting

1. The count refers to a clearly defined amount of time, space or exposure
2. The expected rate of occurrence is constant across comparable parts of the interval
3. Counts in non-overlapping parts of the interval are modelled as independent
4. Over a very small subinterval, the probability of more than one event is negligible relative to the probability of one event

The Poisson Probability Model

The Poisson distribution is determined by one positive parameter, usually written as λ.

The parameter λ is the mean number of occurrences in the specified interval.

$$X \sim \text{Pois}(\lambda)$$

For a given $X \sim \text{Pois}(\lambda)$, then

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

for $k = 0, 1, 2, \ldots$.

The Poisson distribution has the special property that its mean and variance are equal.

$$\mu = \lambda$$

$$\sigma^2 = \lambda$$

$$\sigma = \sqrt{\lambda}$$

Example 12.10: Monitoring Mumps Cases

Question

Suppose the baseline number of mumps cases in a month follows a Poisson distribution with mean $\lambda = 0.1$. What is the probability of observing no more than one case in a month?

Explanation

The phrase *no more than one* includes 0 and 1. These outcomes are disjoint, so their probabilities can be added.

$$P(X \leq 1) = P(X = 0) + P(X = 1)$$

Example 12.10: Monitoring Mumps Cases

Using the Poisson formula,

$$P(X \leq 1) = \frac{e^{-0.1}(0.1)^0}{0!} + \frac{e^{-0.1}(0.1)^1}{1!} = 0.9048 + 0.0905 = 0.9953$$

Under this baseline model, a month with zero or one case occurs about 99.53% of the time.

Example 12.10: Monitoring Mumps Cases

The cumulative Poisson probability can be found with `ppois()` or by adding exact masses from `dpois()`.

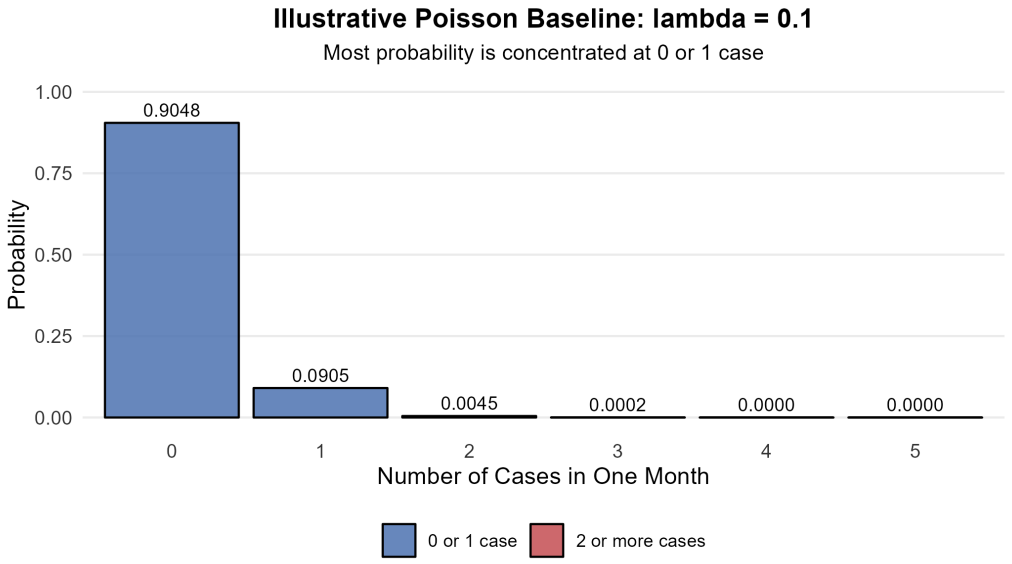
```
# P(X <= 1) for lambda = 0.1
prob_mumps_1_cum <- ppois(q = 1, lambda = 0.1)

prob_mumps_1_manual <-
  dpois(x = 0, lambda = 0.1) + dpois(x = 1, lambda = 0.1)


print(round(prob_mumps_1_cum, 4))
> [1] 0.9953

print(round(prob_mumps_1_manual, 4))
> [1] 0.9953
```

Example 12.10: Monitoring Mumps Cases



Example 12.11: A Count Far Above the Baseline

Question

The same baseline model has $\lambda = 0.1$ case per month. What is the probability of observing four or more cases in one month if the Poisson baseline model remains valid?

Explanation

The desired probability is an upper tail extending from four to infinity. The complement rule turns the problem into a cumulative probability through three cases.

$$P(X \geq 4) = 1 - P(X \leq 3)$$

Example 12.11: A Count Far Above the Baseline

For $X \sim \text{Poisson}(0.1)$, the probability of k cases is

$$P(X = k) = \frac{e^{-0.1}(0.1)^k}{k!}$$

Using the complement rule

$$\begin{aligned} P(X \geq 4) &= 1 - P(X \leq 3) \\ &= 1 - \sum_{k=0}^{3} \frac{e^{-0.1}(0.1)^k}{k!} \\ &= 1 - \left(\frac{e^{-0.1}(0.1)^0}{0!} + \frac{e^{-0.1}(0.1)^1}{1!} + \frac{e^{-0.1}(0.1)^2}{2!} + \frac{e^{-0.1}(0.1)^3}{3!} \right) \\ &\approx 1 - 0.9999961532 \approx \boxed{0.00000385} \end{aligned}$$

Four or more cases would be extremely unusual under this baseline. Such a count may prompt investigation of the rate or the model's assumptions.

Example 12.11: A Count Far Above the Baseline

The upper-tail probability can be calculated with the complement rule or with `lower.tail = FALSE`.

```
# Upper-tail probability P(X >= 4)
prob_outbreak <- 1 - ppois(q = 3, lambda = 0.1)
prob_outbreak_alt <- ppois(q = 3, lambda = 0.1, lower.tail = FALSE)

print(prob_outbreak)
> [1] 3.846834e-06
print(prob_outbreak_alt)
> [1] 3.846834e-06
```

Poisson Probabilities and Surveillance

Example: Monitoring Wildlife Deaths

Suppose a wildlife monitoring programme usually records an average of 2 dead seabirds per week along a fixed coastal route. If deaths occur independently at a stable rate, the weekly count can be modelled as

$$X \sim \text{Poisson}(2)$$

One week, the team records 8 dead seabirds. Under the baseline model

$$P(X \geq 8) \approx 0.0011 = 0.11\%$$

What does this tell us?

A count of 8 or more is therefore unusual under the assumed baseline and may prompt an investigation. The count alone does not identify a cause. Disease, severe weather, or changes in monitoring effort could help explain the increase.

Calculating Poisson Probabilities with R

The R functions for the Poisson distribution follow the same pattern as the binomial functions.

Probability question	R function	Meaning
Exactly x occurrences	`dpois(x, lambda)`	Probability mass at one count
At most q occurrences	`ppois(q, lambda)`	Cumulative probability from 0 through q

The option `lower.tail = FALSE` is useful for upper-tail calculations.

How Should the Model Be Chosen

The model should be selected from the data-generating process.

- A question about a fixed number of patients, seeds, products or policy renewals suggests checking the binomial conditions
- A question about the number of events during a month, within a quadrat or over another fixed exposure suggests checking the Poisson assumptions

A count being a whole number is not enough to determine which model should be used.

Binomial and Poisson Models: Choosing the Right Count Model

Feature	Binomial model	Poisson model
What is counted	Successes among trials	Occurrences in an interval
Upper limit	Fixed at n	No fixed upper limit
Main parameters	n and p	λ
Mean	np	λ
Variance	$np(1-p)$	λ
Exact R function	`dbinom()`	`dpois()`
Cumulative R function	`pbinom()`	`ppois()`